

Aim :- AdaBoost Ensemble Learning

1. Implement AdaBoost algorithm to create an ensemble of weak classifiers.
2. Train the ensemble model on the phishing website dataset and evaluate its performance.
3. Compare the results with an individual weak classifier.

Step	Function / Method Used	Description
1	<code>pd.read_csv()</code>	Loads the phishing website dataset from the CSV file.
2	<code>df.dropna()</code>	Removes rows containing missing values.
3	<code>df.drop_duplicates()</code>	Removes duplicate records from the dataset.
4	Target separation	Separates website features from the <code>status</code> target column.
5	<code>y.map()</code>	Converts legitimate and phishing classes into numeric labels 0 and 1.
6	<code>pd.to_numeric()</code>	Converts feature values into numeric form.
7	<code>train_test_split()</code>	Splits the dataset into 80% training and 20% testing data.

8	<code>DecisionTreeClassifier(max_depth=1)</code>	Creates a Decision Stump as the weak classifier.
9	<code>weak_classifier.fit()</code>	Trains the individual weak classifier.
10	<code>weak_classifier.predict()</code>	Predicts the classes using the weak classifier.
11	<code>AdaBoostClassifier()</code>	Creates the AdaBoost ensemble model.
12	<code>n_estimators=50</code>	Uses 50 weak learners in the AdaBoost ensemble.
13	<code>ada_model.fit()</code>	Trains AdaBoost by focusing more on previously misclassified samples.
14	<code>ada_model.predict()</code>	Predicts legitimate and phishing websites using AdaBoost.
15	<code>accuracy_score()</code>	Calculates and compares the accuracy of both models.
16	<code>classification_report()</code>	Displays precision, recall and F1-score for AdaBoost.
17	<code>confusion_matrix()</code> and <code>plt.imshow()</code>	Creates and visualizes the AdaBoost confusion matrix.
18	<code>plt.bar()</code> and <code>improvement</code>	Compares model accuracies and calculates the improvement from AdaBoost.

CODE:

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T090_PRACTICALS_Alpay - C:\practical\AI\T090_PRACTICALS_Alpay (3.13.0)
File Edit Format Run Options Window Help
import pandas as pd
import matplotlib.pyplot as plt

from sklearn.model_selection import train_test_split
from sklearn.tree import DecisionTreeClassifier
from sklearn.ensemble import AdaBoostClassifier

from sklearn.metrics import accuracy_score
from sklearn.metrics import classification_report
from sklearn.metrics import confusion_matrix

df = pd.read_csv("dataset_phishing.csv")
print("Dataset loaded successfully!")

print("\nOriginal Dataset Shape:")
print(df.shape)

df = df.dropna()
print("\nMissing values removed successfully!")

df = df.drop_duplicates()
print("\nDataset Shape After Removing Duplicates:")
print(df.shape)

target_column = "status"
print("\nTarget Column:")
print(target_column)

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X = df.drop([target_column, "url"], axis=1)
y = df[target_column]

y = y.map({
    "legitimate": 0,
    "phishing": 1
})

X = pd.to_numeric(X.stack(), errors="coerce").unstack()

valid_rows = y.notna() & X.notna().all(axis=1)

X = X.loc[valid_rows]
y = y.loc[valid_rows]

print("\nNumber of Features:")
print(X.shape[1])

print("\nTarget Classes:")
print(y.value_counts())

X_train, X_test, y_train, y_test = train_test_split(
    X,
    y,
    test_size=0.20,
    random_state=42,
    stratify=y
)

print("\nTraining Samples:")
print(len(X_train))

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T090_PRACTICAL6_Alpay - C:\practical\AI\T090_PRACTICAL6_Alpay (3.13.0)
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print("\nTesting Samples:")
print(len(X_test))

weak_classifier = DecisionTreeClassifier(
    max_depth=1,
    random_state=42
)

weak_classifier.fit(X_train, y_train)

weak_prediction = weak_classifier.predict(X_test)

weak_accuracy = accuracy_score(
    y_test,
    weak_prediction
)

print("\nWeak Classifier Accuracy:")
print(round(weak_accuracy * 100, 2), "%")

ada_model = AdaBoostClassifier(
    n_estimators=50,
    learning_rate=1.0,
    random_state=42
)

ada_model.fit(X_train, y_train)

ada_prediction = ada_model.predict(X_test)

ada_accuracy = accuracy_score(
    y_test,
    ada_prediction
)

print("\nAdaBoost Accuracy:")
print(round(ada_accuracy * 100, 2), "%")

print("\nClassification Report:")
print(classification_report(
    y_test,
    ada_prediction,
    target_names=["Legitimate", "Phishing"],
    zero_division=0
))

cm = confusion_matrix(
    y_test,
    ada_prediction
)

print("\nConfusion Matrix:")
print(cm)

plt.figure(figsize=(7, 6))

plt.imshow(cm)

plt.title("AdaBoost Confusion Matrix")
plt.xlabel("Predicted Class")
plt.ylabel("Actual Class")

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T090_PRACTICAL6_Alpay - C:\practical\AI\T090_PRACTICAL6_Alpay (3.13.0)
File Edit Format Run Options Window Help
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File Edit Format Run Options Window Help
plt.xticks(
    [0, 1],
    ["Legitimate", "Phishing"]
)

plt.yticks(
    [0, 1],
    ["Legitimate", "Phishing"]
)

for i in range(2):
    for j in range(2):
        plt.text(
            j,
            i,
            str(cm[i, j]),
            ha="center",
            va="center"
        )

plt.colorbar()
plt.tight_layout()

plt.show()

models = [
    "Weak Classifier",
    "AdaBoost"
]

accuracies = [
    weak_accuracy * 100,

```

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T090_PRACTICAL6_Alpy - C:\practical\AI\T090_PRACTICAL6_Alpy (3.13.0)
File Edit Format Run Options Window Help
plt.figure(figsize=(8, 6))

plt.bar(
    models,
    accuracies
)

plt.title("Weak Classifier vs AdaBoost")
plt.xlabel("Model")
plt.ylabel("Accuracy (%)")

for i, value in enumerate(accuracies):
    plt.text(
        i,
        value + 1,
        f"{value:.2f}%",
        ha="center"
    )

plt.ylim(0, 100)
plt.tight_layout()

plt.show()

improvement = (
    ada_accuracy - weak_accuracy
) * 100

print("\nAccuracy Improvement:")
print(round(improvement, 2), "%")

print("\n=====")

```

```

T090_PRACTICAL6_AI.py - C:/practical/AI/T090_PRACTICAL6_AI.py (3.13.0)
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plt.text(
    i,
    value + 1,
    f"{value:.2f}%",
    ha="center"
)

plt.ylim(0, 100)
plt.tight_layout()

plt.show()

improvement = (
    ada_accuracy - weak_accuracy
) * 100

print("\nAccuracy Improvement:")
print(round(improvement, 2), "%")

print("\n=====")
print("ADABOOST PRACTICAL COMPLETED")
print("=====")

print("\nWeak Classifier Accuracy:",
      round(weak_accuracy * 100, 2), "%")

print("\nAdaBoost Accuracy:",
      round(ada_accuracy * 100, 2), "%")

print("\nImprovement:",
      round(improvement, 2), "%")

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OUTPUT:

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IDLE Shell 3.13.0*
File Edit Shell Debug Options Window Help

Python 3.13.0 (tags/v3.13.0:60403a5, Oct 7 2024, 09:38:07) [MSC v.1941 64 bit (AMD64)] on win32
Type "help", "copyright", "credits" or "license()" for more information.
>>>
===== RESTART: C:/practical/AI/T090_PRACTICAL6_AI.py =====
Dataset loaded successfully!

Original Dataset Shape:
(11430, 89)

Missing values removed successfully!

Dataset Shape After Removing Duplicates:
(11430, 89)

Target Column:
status

Number of Features:
87

Target Classes:
status
0    5715
1    5715
Name: count, dtype: int64

Training Samples:
9144

Testing Samples:
2286

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status
0    5715
1    5715
Name: count, dtype: int64

Training Samples:
9144

Testing Samples:
2286

Weak Classifier Accuracy:
85.17 %

AdaBoost Accuracy:
92.78 %

Classification Report:
              precision    recall  f1-score   support

 Legitimate      0.93        0.93        0.93        1143
  Phishing      0.93        0.93        0.93        1143

 accuracy      0.93        0.93        0.93        2286
 macro avg      0.93        0.93        0.93        2286
 weighted avg      0.93        0.93        0.93        2286

Confusion Matrix:
[[1058   85]
 [  80 1063]]
```

GRAPH:



